

Machine Learning Suggestion

Module 1:

Introduction:

1. What is machine learning? Explain briefly the different types of machine learning.
2. How is supervised learning different from unsupervised learning?
3. How is regression different from classification?
4. What do you mean by overfitting and underfitting of a model?
5. What is cost function in machine learning?
6. What is training set and test set in machine learning model?
7. What is the trade-off between an Overfitting and underfitting model in regression modeling?

Supervised Learning:

Linear Regression:

1. What is a linear regression algorithm? How do you interpret a linear regression model?
2. How does linear regression work?
3. What are the assumptions of linear regression?
4. Why are they important?

5. Numerical question

Logistic Regression:

1. What is Logistic Regression?
2. What is Logistic Function or Sigmoid Function?

3. What is the range of values of a logistic function?
4. What is the odds ratio?
5. Why does logistic regression come under the category of classification problems?
6. How is logistic regression different from linear regression?

KNN:

1. Which machine learning algorithm is called lazy learner and why?
2. Write the advantages and disadvantages of KNN algorithm.
3. Why is KNN a non-parametric Algorithm?
4. Why is the odd value of "K" preferred over even values in the KNN algorithm?
5. How does the KNN algorithm make the predictions on the unseen dataset?

6.Numerical question

7. What is the space and time complexity of the KNN Algorithm?
8. Can the KNN algorithm be used for regression problem statements?
9. Write the advantages and disadvantages of KNN algorithm.

Naïve Bayes Classifier:

1. Define Bayes theorem.
2. What is the basic assumption in Naïve Bayes classifier?

3.Numerical question

4. What is conditional probability and why is it important to understand when working with Naïve Bayes?
5. What is the basic assumption in the case of the Naïve Bayes classifier?
6. Explain how Naïve Bayes classifier handles continuous data attributes.

Decision Tree:

- 1.Explain the terms Information Gain and Gini index with the help of an example.
- 2.Construct a decision tree Training data set is given.
- 3.Numerical question**
- 4.What is a decision tree algorithm? Explain with an example.

SVM:

- 1.Explain the working principle of support vector machine algorithm.
- 2.Write the difference between linear and non-linear SVM.
- 3.Find the optimal hyper plane for the set of following data points.
- 4.What is the Euclidian distance between two data points?
- 5.Numerical question (from linear SVM)**
6. How does SVM handle non-linear classification problems? Illustrate with a diagram.

Confusion Matrix:

- 1.Confusion matrix will be given. Find out the accuracy, misclassification rate and sensitivity.
- 2.What are True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) in a confusion matrix?
- 3.Explain the difference between Precision and Recall using confusion matrix terms.
- 4.What is a Confusion Matrix?
- 5.Why Do We Need a Confusion Matrix?
- 6.Numerical question (from Confusion Matrix)**

Module 2:

Unsupervised Learning:

1. Write the differences between supervised and unsupervised learning.
2. Write down the differences between classification and clustering.
3. What do you mean by clustering?
4. Write note on the different types of clustering.

k-means algorithm:

1. Explain briefly the working principles of k-means algorithm.
2. What are the differences between K-means clustering and K-nearest neighbor (KNN).
3. **Numerical question.**
4. How to decide the optimal number of K in the K means Algorithm?
5. What are the advantages and disadvantages of the K means Algorithm?
6. Write the working principle of the K-means algorithm.
7. Use k-means algorithm to evaluate clusters in given data.
8. How does K-Medoids differ from K-Means in choosing cluster centers?
9. How to determine K using the ELbow method?
10. Differentiate between K-means clustering and K-nearest neighbor (KNN).

PCA:

- 1.Explain the term “Curse of Dimensionality”.
- 2.What are the ways of dealing with sparse features?
- 3.What are the common problems with sparse dataset in machine learning?
- 4.Why is machine learning difficult with sparse features?
- 5.What is difference between sparse data and missing data?
- 6.What is principal component analysis?
- 7.What does a PCA do?
- 8.What are the steps involved in PCA algorithm.
- 9.What are the advantages and disadvantages of PCA algorithm?
10. How does PCA transform correlated features into uncorrelated ones?
- 11.What are the benefits and limitations of the PCA algorithm?
- 12.Why is it necessary to standardize data before applying PCA?
- 13.Describe the role of eigenvalues and eigenvectors in PCA.

Matrix Factorization:

- 1.Write a short note on matrix factorization.
- 2.Where Matrix Factorization is used?

Module 3:

- 1.What is meant by Ensembling?
- 2.What is the need for Ensembling multiple learners?
- 3.What do you mean by Ensemble learning? Explain.
- 4.What are the advantages and disadvantages of ensemble learning?
- 5.How can we classify the various ensembles learning technique?
- 6.Compare bagging and boosting.
- 7.What is the principle behind the Random Forest ensemble model?

- 8.What are the advantages and disadvantages of the Random Forest algorithm?
- 9.Explain the differences between Random Forest and Decision Tree?
- 10.What is the general idea of boosting technique?
- 11.Compare AdaBoost and Gradient Boosting.
- 12.What features are added to XGBoost to make it perform better than other boosting algorithms?
- 13.Explain the different advanced ensemble techniques.
- 14.Why do we need Ensemble Learning in machine learning?
- 15.Define base learner in the context of ensembles.
- 16.What is Bagging in ensemble learning?
- 17.Compare AdaBoost and Gradient Boosting
- 18.What is Random Forest algorithm?
- 19.Why are Random Forests algorithm less prone to overfitting than individual trees?

Module 4:

- 1.State the difference between Semi-supervised and Reinforcement learning.
2. What are the limitations of semi supervised learning?
- 3.Define seasonal index.
- 4.Explain cyclic variations.
- 5.Define Time series and time series data.
- 6.What are the different methods for time series forecasting?
- 7.Explain with examples the different component of time series.
- 8.Explain the architecture of an artificial neural network (ANN).
- 9.What is the role of the Activation functions in Neural Networks?